REPORTS

The 945 TWh Question: Why AI's Power Crisis Is the Investment Story Most Investors Are Still Missing

July 30, 2026

> *The IEA's Energy and AI special report — 304 pages, published by the institution that defines the global energy outlook — has put a specific number on something most AI infrastructure coverage has been talking around. Data centre electricity consumption more than doubles to 945 terawatt-hours by 2030. That number reorders the entire AI infrastructure investment thesis. This is the constraint I had been pricing wrong.*

17 min · Reports

Why AI's Power Crisis Is the Investment Story Most Investors Are Still Missing

The International Energy Agency does not write hype. That is what makes its *Energy and AI* special report — a 304-page analysis published in 2025 and updated in early 2026, peer-reviewed by Google, Microsoft, Amazon Web Services, OpenAI, and most major utilities — the most important AI infrastructure document I have read in months.

It put a specific number on something most of the AI investment conversation has been talking around without quite saying out loud.

Global data centre electricity consumption goes from approximately 415 terawatt-hours in 2024 to around 945 terawatt-hours by 2030. That is more than a doubling in six years, in a category that has historically grown at roughly the same pace as global GDP. By 2035, the IEA's base case has the same demand reaching around 1,200 terawatt-hours — more than the total electricity consumption of Japan today.

That number is the constraint I had been pricing wrong. This piece is about why, what the IEA data tells us that the equity market is still digesting in installments, and which layer of the power stack the asymmetric trades concentrate in.

---

1. What I had been missing

I want to be honest about the framing I had been running.

For most of the AI infrastructure conversation through 2024 and into 2025, I treated power as the macro overlay. Compute was the asset. Memory was the supply constraint. Networking was the plumbing. Power was the background condition — important, but in the same way that "the economy" or "interest rates" are important. It mattered everywhere. It mattered specifically nowhere.

That framing was wrong, and the IEA data is what made the specific shape of how it was wrong clear to me.

Power is not the macro overlay of AI infrastructure. It is, increasingly, the binding constraint on how fast the entire system can scale. The reason most investors are still under-pricing this is structural: the institutions that allocate the most aggressively into AI exposure — large funds, hyperscaler equity holders, semiconductor specialists — are mostly not power-sector specialists. The institutions that understand the power sector best — utility investors, energy infrastructure funds, grid operators — are mostly not at the front of the AI conversation. The result is a category of trades that sit in the gap between those two communities, and the IEA report essentially names that gap explicitly.

This piece is my attempt to think about the trades that sit in that gap.

---

2. The headline numbers — and why each one matters

Before getting into the framework, I want to set the data out cleanly. These are the numbers from the IEA report that I have been turning over for the last few weeks, in roughly the order of importance for the investment thesis.

415 TWh in 2024, growing to ~945 TWh by 2030. Data centres consumed approximately 1.5 percent of global electricity in 2024. That share is on track to roughly double by 2030. The United States accounted for 45 percent of global data centre electricity consumption in 2024, China 25 percent, Europe 15 percent. By 2030, US data centres alone are projected to consume more electricity than US production of aluminium, steel, cement, chemicals, and all other energy-intensive goods combined.

Data centre demand grew at roughly 12 percent annually since 2017 — more than four times faster than total electricity demand growth over the same period. The category is no longer a small contributor inside a broader trend. It is one of the dominant drivers of accelerated electricity demand growth in advanced economies, where total electricity consumption has been essentially flat for the better part of two decades.

Half of US data centre capacity sits in five regional clusters. The geographic concentration is the part the equity market has the most trouble pricing because it does not show up in national-level data. It shows up in transformer queues in Virginia, transmission constraints in Texas, water and power siting battles in Arizona, and increasingly in interconnection backlogs that delay specific named projects by years rather than months.

An AI-focused data centre is approximately ten times more capital-intensive per kilowatt than an aluminium smelter and roughly fifty times more capital-intensive than an air conditioner. Investment costs can reach $30,000 per kilowatt for accelerated computing servers. That number reframes what data centres are, economically. They are not industrial loads. They are infrastructure assets whose cost structure rewards near-full utilization, which in turn means they cannot easily be operated flexibly to balance the grid.

Roughly 20 percent of planned data centre projects are at risk of delays absent decisive action on grid bottlenecks. Building new transmission lines takes four to eight years in advanced economies. Wait times for transformers and cables have doubled in the past three years. Gas turbine delivery lead times now extend several years for new gas-fired plants — meaning commissioning gets pushed past 2030 for projects that look fully financed and permitted today.

Cumulative data centre investment 2025-2030 in the IEA base case: $4.2 trillion in IT and data centre capex, plus approximately $480 billion in energy capex. Annual investment was approximately $100 billion in 2015. It grew to over $500 billion in 2024. The IEA expects it to surpass $800 billion per annum before 2030.

Those are the numbers. Each one is a separate pricing question for the public market. Together, they describe a category transition that is too large to be priced as a normal capital expenditure cycle.

---

3. The framework — three layers of the power stack

The cleanest way I have found to think about the power investment opportunity is to apply the same three-layer structure I have been using for memory, storage, and optical networking. Power, like those other layers, has different dynamics at different parts of the stack.

Layer 1 — Generation. This is the layer that produces the electrons. Nuclear (both existing fleet and small modular reactors), natural gas (combined-cycle and peaking), renewables paired with storage, and emerging sources like geothermal. The IEA base case has renewables covering roughly half of incremental data centre demand to 2035, with natural gas providing about 175 TWh of additional generation, nuclear contributing approximately the same amount, and the first small modular reactors coming online around 2030.

Layer 2 — Grid and Equipment. This is the layer that moves the electrons. Transmission lines, distribution infrastructure, substations, transformers, and the specialized components — turbines, switchgear, cable — that the rest of the system depends on. This is the layer where the IEA's "20 percent of projects at risk of delay" number is concentrated. Building new transmission takes years. Transformer lead times have doubled. The grid is the chokepoint inside the power chokepoint.

Layer 3 — Data Centre Integration. This is the layer that consumes the electrons and the layer where the AI capex is most visible. Hyperscaler-owned facilities, third-party colocation operators, the increasing share of capacity being built specifically for AI training and inference workloads. This is the layer the equity market understands best, and probably the layer where the easy money has already been priced.

The framework matters because the rerating dynamics are different at each layer. Layer 3 is well into its repricing. Layer 1 is mid-cycle and uneven across generation sources. Layer 2 is barely being priced at all in most public comparables, even though the IEA data explicitly identifies it as the binding constraint.

That asymmetry is where the investment work concentrates.

---

4. Layer 1 — Generation, by source

Layer 1 is the most diversified layer of the power stack, and the one where the investment thesis splits most clearly along source-specific lines.

Nuclear is, in my view, the most under-priced generation source relative to the structural demand growth the IEA describes. The legacy operators — Constellation Energy in the US, Cameco for upstream fuel cycle, BWX Technologies for small modular reactor exposure — sit on assets that produce dispatchable, near-zero-emissions baseload power at a moment when hyperscalers are openly underwriting long-term power purchase agreements. The Microsoft-Three Mile Island restart is the most visible single example, but the broader pattern is more important: utilities with existing nuclear capacity are negotiating contracts at terms that price the supply as scarce, not abundant. The IEA's base case for nuclear contribution to data centre demand growth — roughly 175 TWh by 2035 — is a meaningful upgrade to the asset class's pricing thesis. The first commercial small modular reactors arriving around 2030 extend the runway further.

Natural gas captures the largest single-source contribution to incremental data centre demand in the IEA base case: roughly 175 TWh of additional generation by 2035, concentrated in the United States. The investment implications run in two directions. On the equipment side, GE Vernova, Siemens Energy, and Mitsubishi Heavy Industries (through MHI's gas turbine business) are positioned where the supply constraint is most acute — turbine lead times of several years now sit between fully financed projects and commercial operation. On the operations side, US-based gas-fired power generators (Vistra, NRG, the regulated utility operators of combined-cycle capacity) are picking up new long-duration demand contracts at favorable terms. The thesis is not that gas is the long-term winner. It is that gas is the binding-constraint solution in the time window before nuclear and renewables-plus-storage scale fully into place.

Renewables and storage cover the largest share of total incremental demand in the IEA base case — roughly 450 TWh of additional renewable generation to meet data centre demand by 2035, supported by storage and the broader grid. The first-order investment exposure is to the existing utility-scale solar, wind, and storage developers. The IEA's analysis of 99-percent hourly matching with renewables-plus-storage portfolios is, technically, achievable for hyperscaler-scale loads. The pricing is the binding constraint. Where 24/7 carbon-free matching is required (Google, Microsoft commitments), the cost structure tilts toward hybrid portfolios that pair renewables with dispatchable sources.

Geothermal and other emerging sources are the longer-duration option. The hyperscaler-driven investment in next-generation geothermal — Fervo, Sage, and several private developers — is one of the cleaner examples of tech-sector capital underwriting an energy-sector technology curve that the broader equity market is still mostly ignoring.

The investable read on Layer 1 is not "buy power." It is that the relative pricing across generation sources is being reshuffled by an underlying demand shift that none of the historical valuation models for utilities were calibrated against. Specific exposure decisions matter more than category exposure.

---

5. Layer 2 — The grid is the binding constraint

If Layer 1 is the most diversified part of the power stack, Layer 2 is the most concentrated — and the most binding.

The IEA's framing of grid constraints is unusually direct. New transmission line construction in advanced economies now takes four to eight years from initial proposal to commercial operation. Transformer and high-voltage cable lead times have doubled in three years. Interconnection queues for data centres, renewables, and other large new loads have grown to multi-year backlogs in major North American markets. The combination is a physical limit on how fast the AI infrastructure capex cycle can deploy regardless of how much equity or debt is available to fund it.

The investment implication is that the companies producing the bottleneck equipment — high-voltage transformers, switchgear, cable, transmission components — are operating in a market with explicit, structural supply constraints. Hitachi Energy (recently spun out of ABB into focus), Siemens Energy's grid business, GE Vernova's transmission and distribution segment, Eaton, and the specialized cable producers (Prysmian, NKT) are the cleanest direct exposures. The thesis is similar in shape to the laser-and-EML chokepoint I have been describing for optical networking — a small number of suppliers producing a critical component category whose lead times are extending faster than capacity additions can offset.

The regional pattern matters too. The IEA's identification of geographic clustering — particularly the five US regional clusters that hold roughly half of national data centre capacity — implies that the next phase of buildout will increasingly need to occur in regions with less existing infrastructure. That, in turn, increases the demand for new transmission, new substations, and new grid integration work in places that have not seen meaningful transmission investment in years. The companies positioned to do that work, on accelerated timelines, are the same Layer 2 names whose pricing is starting to reflect the durability of the demand.

This is the layer I think is most consistently mispriced in the current equity market. It is too unglamorous to attract the AI capex narrative directly. It is too structurally exposed to the AI demand profile to be priced as a normal utility-equipment cycle. The asymmetry is what makes it interesting.

---

6. Layer 3 — Data centre operators and the visible AI exposure

Layer 3 is the layer the equity market is most comfortable with: the data centre operators themselves. Equinix and Digital Realty as the largest public colocation operators, the hyperscalers as the largest in-house operators, and a long tail of specialized AI-focused developers building purpose-built capacity for training and inference workloads.

The IEA data does two things to the existing investment narrative here.

First, it validates the underlying demand story in a way that internal hyperscaler commentary alone has not been able to. The 945 TWh number is institution-grade evidence that the demand profile most operators have been describing privately is real, durable, and broadly understood at the macroeconomic level. That validation matters for valuation multiples that have been carrying skepticism about whether the AI capex cycle is sustainable.

Second, it reframes how to think about geographic exposure. The same five-cluster concentration that creates grid risk also creates pricing power for existing operators in those markets. The combination of (a) durable demand growth, (b) constrained new supply because of grid and equipment bottlenecks, and (c) increasing premium for AI-specific capacity gives operators in primary markets the ability to push power-pricing pass-through and inflation-indexing terms that did not exist in the same way three years ago.

The honest constraint here is that the easy money in Layer 3 has already been priced. EQIX and DLR have meaningfully outperformed broad market indices through the AI capex cycle. The asymmetry is no longer obvious. What remains is the question of whether the AI-specific operators — the ones building purpose-built capacity for training workloads, the ones securing power capacity through hyperscaler-style PPAs, and the ones with credible roadmaps for AI-optimized cooling and density — get rerated separately from the broader colocation category.

That separation is the part the equity market is still working out.

7. The geographic dimension

The IEA report does something most equity research on AI capex does not: it takes the geography seriously.

The five US regional clusters — most prominently Northern Virginia, but also clusters in Dallas-Fort Worth, Phoenix, Chicago/Northern Illinois, and Atlanta — hold roughly half of US data centre capacity. The next-largest concentrations sit in specific Chinese provinces, Western Europe (particularly Ireland and the Netherlands), and a smaller set of emerging clusters in Southeast Asia and the Nordics.

The investment implications are concrete. Utilities operating in those regions are seeing a class of demand growth that most other US utilities are not. Dominion Energy's exposure to Northern Virginia. American Electric Power's exposure to the Ohio Valley cluster. Vistra's gas generation portfolio across Texas. Southern Company's nuclear and gas mix across the Southeast. These are not generic utility exposures. They are concentrated, AI-correlated utility exposures whose forward earnings power depends on whether the local grid and generation mix can keep pace with new load.

The IEA's analysis of 50 percent of US data centres under development being located in pre-existing large clusters is the warning bell. Local grids in those clusters are already constrained. The next phase of buildout in those locations will require either grid investment that takes years, or relocation to less-constrained regions that takes operational re-planning. Either way, the pricing of grid access and power supply in those clusters is moving from "abundant utility input" to "constrained strategic input."

For investors, the regional read changes the investment frame from sector-level utility exposure to specific-cluster utility exposure. The latter is a more concentrated bet with a clearer underlying thesis.

8. The critical-minerals dimension

The IEA report flagged one more constraint that I think most AI infrastructure investors are still missing: the upstream critical-minerals dependence underneath the entire data centre supply chain.

Gallium — used in cutting-edge computer chips and power electronics — is one example. China currently accounts for approximately 99 percent of global refined gallium supply. The IEA estimates that gallium demand from data centres alone could reach over 10 percent of today's total supply by 2030.

That is not an isolated case. Similar concentration patterns exist across the critical-mineral inputs for batteries (lithium, cobalt), power electronics (rare earth permanent magnets), and grid equipment (copper, aluminium, specialty steels). The supply chain underneath AI infrastructure is more geographically concentrated than the supply chain underneath the last industrial cycle, and it is more politically exposed to trade-policy outcomes that are increasingly contested in 2026.

The investable implication is twofold. First, the small group of non-Chinese producers of these critical minerals are seeing strategic-input pricing dynamics similar to what I described for lasers and EMLs in the optical stack — concentrated supply, structural demand, and customer behavior consistent with long-duration underwriting rather than spot procurement. Second, the policy and trade-flow risk attached to AI infrastructure capex is meaningfully larger than the equity market currently prices for most operators, and the companies with credible non-China supply chains will trade at premiums to those that do not as that risk surfaces.

This dimension deserves its own piece. The IEA framing tells me it is not yet in the equity research mainstream, and that is usually where I have found the most useful work to do.

9. Where the thesis breaks — the bear cases I take seriously

Any thesis this structural has to be paired with a serious bear case. There are four I think are worth owning honestly.

Efficiency outpaces demand. The IEA's High Efficiency Case projects that data centre electricity demand could be 20 percent lower in 2035 than in the base case if AI hardware and model efficiency improve faster than the base trajectory. If next-generation accelerators reduce per-token power consumption materially, or if model architectures become significantly more memory- and compute-efficient, the demand curve flattens. The base case is not guaranteed. The Headwinds Case is also explicitly modeled.

Capex digestion and macro slowdown. The AI capex cycle has been running at a pace that most analysts agree cannot continue indefinitely. A capex digestion phase — driven by recession, hyperscaler discipline, or simply the natural slowing of a multi-year buildout — would push out the demand profile and re-test the structural pricing of generation, grid, and operator names. The framework predicts the structural thesis survives a normal cycle. A severe cycle would test that prediction harder than the current data has.

Technology substitution. If alternative compute architectures (efficiency-focused custom silicon, distributed edge AI, fundamentally different training paradigms) reduce the data-centre-centric profile of AI workloads, the power demand profile changes shape. This is the lowest-probability bear case in the near term — accelerated compute concentration in large facilities is the visible trend — but it is not zero-probability over a decade-long horizon.

Policy fragmentation and trade-flow shocks. The geographic concentration of both demand (the five US clusters) and supply (the critical-minerals chain) means the asset class is more exposed to policy shocks than the broader equity market typically prices for it. A sharp escalation in US-China technology decoupling, an aggressive export-control regime targeting specific data centre components, or a trade-policy outcome that disrupts the renewables supply chain could compress the demand profile faster than any cyclical risk could.

Owning the bull case responsibly means owning these risks. The structural shift is real. The conditions that allow it to compound are not all under the operators' control.

---

10. What this means for U.S. investors

The practical takeaway is a positioning framework, not a single-stock call.

The three-layer structure of the power stack suggests three different ways to express the same broad thesis, with different risk profiles and different timelines.

Layer 1 exposure means generation assets, with the relative pricing across nuclear, gas, renewables, and emerging sources being repriced by hyperscaler underwriting. Nuclear (Constellation, BWXT, Cameco) is the most asymmetric of the major generation sources in my read. Natural gas equipment (GE Vernova, Siemens Energy) captures the constrained-equipment leg of the buildout. Utility-scale renewables and storage continue to grow into the largest single share of incremental supply.

Layer 2 exposure means grid and equipment, where the supply constraint is most binding and where the equity market is most under-pricing the structural component. Hitachi Energy, Siemens Energy grid business, GE Vernova transmission and distribution, Eaton, and the specialty cable producers are the cleanest direct exposures. This is the layer where I think the most consistently mispriced trades sit.

Layer 3 exposure means data centre operators and the AI-correlated utilities. EQIX, DLR, and the regional utilities with concentrated AI-cluster exposure (Dominion, AEP, Southern, Vistra) carry validated demand profiles but increasingly priced multiples. The asymmetry sits less in the category exposure and more in the differentiation between general colocation and AI-purpose-built capacity.

The framework does not say that all three layers will be equally rewarding. It says they will be rewarding differently, on different timelines, with different downside profiles. Investors who treat power as one undifferentiated category will likely over-pay for the well-priced operator names and under-own the constrained-equipment layer where the real structural pricing power sits.

The framework also does not say that "power is going up from here." It says the asset class is being recategorized from cyclical utility exposure to AI-correlated infrastructure exposure, the rerating is uneven across layers, and the IEA data has given us institution-grade evidence that the demand profile is real and durable enough to justify the structural framing.

Closing

The IEA does not write hype. When the institution that defines the global energy outlook publishes a 304-page report putting specific numbers on AI's electricity demand, the constraint moves from speculative narrative to institution-grade fact.

The 945 TWh number is the part of the report that will get cited most. It deserves the attention. But the more important content sits in the surrounding analysis — the geographic concentration, the equipment lead times, the critical-minerals supply chain, the ten-times capital intensity differential versus traditional industrial loads. Each one is a constraint that an AI infrastructure thesis built only on compute and memory does not adequately price.

The compute layer gets the headlines. The memory layer is being repriced. The optical layer is the next chokepoint. And the power layer — generation, grid, and integration — is the constraint underneath all of them.

If I had to compress this report into one sentence, it would be this:

AI's capex cycle is no longer compute-bound or memory-bound. It is increasingly power-bound, grid-bound, and minerals-bound — and the equity market is still pricing those constraints in installments.

The next eighteen months will tell us how much of the framework was right and how fast the rerating completes. The grid layer is where I expect the most asymmetric outcomes. The generation layer is where I expect the cleanest source-specific differentiation. The operator layer is where the rerating is most complete.

This is a long-duration thesis with an active capex cycle running underneath it. That combination — structural shift plus immediate revenue tailwind — is the rarer setup. The IEA has put the data on the table. The investment work is figuring out which layer of the stack the exposure actually sits at.

Brutal Edge. Frameworks over forecasts. Signal over noise.